

Topic 5.1

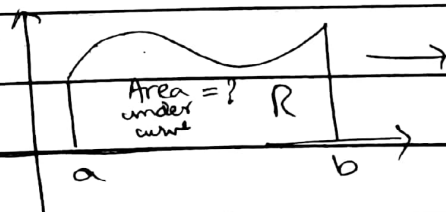
Area & Estimating with Finite Sums

* Estimating with Finite Sums:

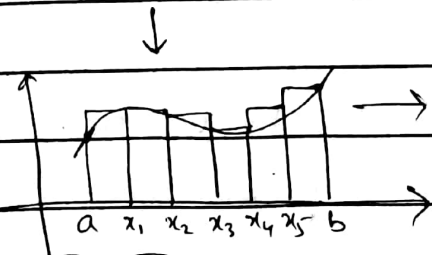
This section shows how area, average values & the distance traveled by an object over time all be approximated by finite sums

* Area:

The area of a region with a curved boundary can be approximated by summing the areas of a collection of rectangles.



R can not be found by simple formula



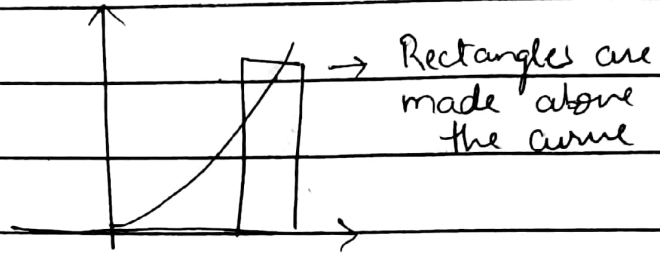
For this we will use approximating rectangles to estimate the area

Area under curve = Sum of the areas of all rectangles.

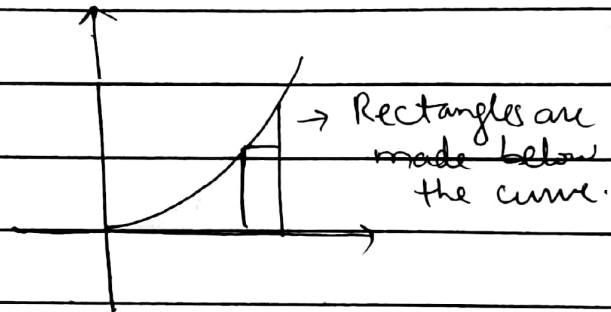
* Method

- (1) Upper Sums
- (2) Lower Sums
- (3) Midpoint Sums

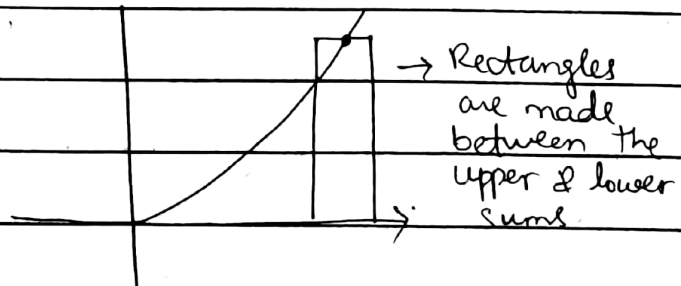
Upper Sum



Lower Sum

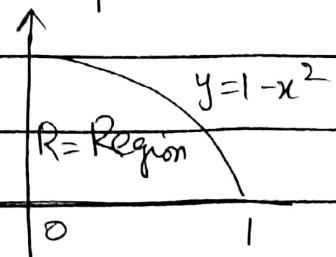


Midpoint Sum



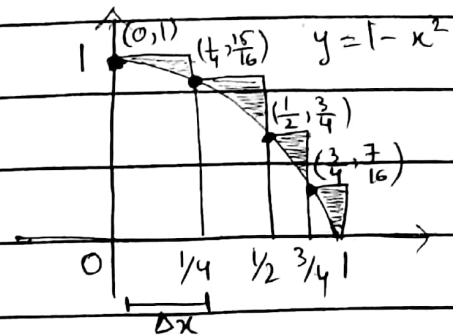
* Example 1 $f(x) = 1 - x^2$ between $x=0$ & $x=1$.

Use finite approximations to estimate the area under the graph of the function using upper, lower & midpoint sums with four rectangles of equal width.



Sol: * Upper Sum:

Since f is decreasing on $[0, 1]$, we use Left ~~Right~~ endpoints to obtain upper sums.



Width $= \Delta x = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}$
 b/t two points of the interval

n = no. of rectangles
 a = initial point of interval
 b = final point of interval.

A = Sum of areas of all rectangles.

$$A = \sum_{k=1}^n f(x_k) \Delta x$$

$$A = f(x_1) \Delta x + f(x_2) \Delta x + f(x_3) \Delta x + f(x_4) \Delta x$$

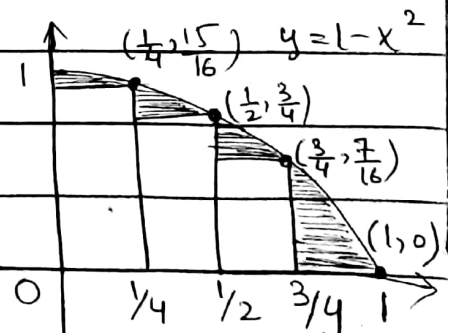
$$A \approx \frac{1 \cdot 1}{4} + \frac{15 \cdot 1}{16 \cdot 4} + \frac{3 \cdot 1}{4 \cdot 4} + \frac{7 \cdot 1}{16 \cdot 4}$$

$$A \approx \frac{25}{32} \approx \boxed{0.78125}$$

* Lower Sum:

Since f is decreasing on $[0, 1]$, we use Right endpoints to obtain lower sums.

$$\Delta x = \frac{1}{4} \text{ (Same as above)}$$



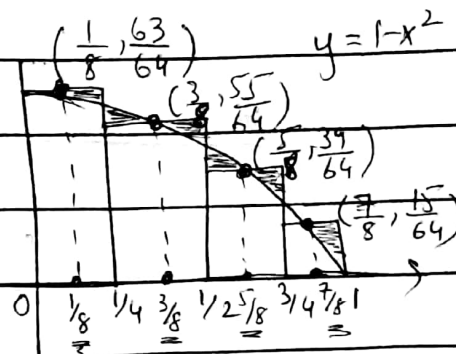
$$A = f(x_1) \Delta x + f(x_2) \Delta x + f(x_3) \Delta x + f(x_4) \Delta x$$

$$\approx \frac{15 \cdot 1}{16 \cdot 4} + \frac{3 \cdot 1}{4 \cdot 4} + \frac{7 \cdot 1}{16 \cdot 4} + \frac{0 \cdot 1}{4} = \frac{17}{32} \approx \boxed{0.53125}$$

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* Midpoint Sum

$$\text{Midpoint} = \frac{a+b}{2} = \frac{0 + \frac{1}{4}}{2} = \frac{\frac{1}{4}}{2} = \frac{1}{8}$$



$$\Delta x = \frac{1}{4}$$

$$\begin{aligned} A &= f(x_1) \Delta x + f(x_2) \Delta x + f(x_3) \Delta x + f(x_4) \Delta x \\ &= \frac{63}{64} \cdot \frac{1}{4} + \frac{55}{64} \cdot \frac{1}{4} + \frac{39}{64} \cdot \frac{1}{4} + \frac{15}{64} \cdot \frac{1}{4} \\ &= \frac{172}{64} \cdot \frac{1}{4} \approx \boxed{0.671875} \end{aligned}$$

• General Formula To Find Area of Curve

$$* A = \sum_{k=1}^n f(x_k) \Delta x \rightarrow \text{Area Approximation Based on 'k'}$$

$\nwarrow$ $\downarrow$ $\downarrow$
 No. of subintervals to be made Height of Rectangle Width of rectangle

* How can we make it better?

By taking $n \rightarrow \infty$

* By definition Formula of Area

$$* A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k) \Delta x$$

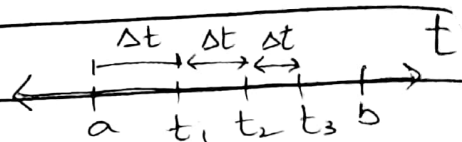
$\nwarrow$ $\downarrow$ $\downarrow$
 Sum function change

* Direct formula of Area

$$* A = \int_a^b f(x) dx \rightarrow \text{change}$$

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* Distance Traveled



Distance = Velocity $\times$ Time, $t \in [a, b]$.

$$D = V(t) \times \Delta t.$$

The sum of distances traveled over all the time intervals = $D \approx V(t_1)\Delta t + V(t_2)\Delta t + \dots + V(t_n)\Delta t$

where n is total no. of subintervals.

→ This sum is only an approximation to the true distance D , but the approximation increases in accuracy as we take more and more subintervals.

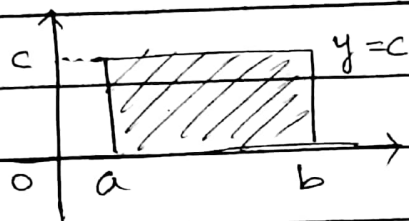
* Example 1 (From book) → (Old Notes)

* Average Value of a Non-negative Cont. functions :

① The average value of a constant function

$$f(x) = c \quad \text{on } [a, b]$$

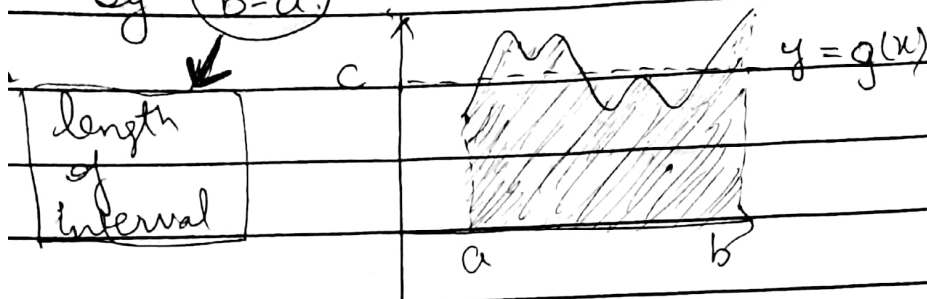
is the area of rectangle divided by $b-a$.



$$\text{Average value of } f(x) = \frac{\text{Length} \times \text{Width}}{b-a}.$$

$$= \frac{c(b-a)}{b-a} = c$$

② The average value of a non-constant function / non-negative function $g(x)$ on an interval $[a, b]$ is the area under/beneath its graph divided by $(b-a)$.



Average value of $g(x)$ over $[a, b]$		$=$	$\frac{\text{Area under curve}}{b-a}$
$\bar{g}$		$=$	$\frac{\sum_{k=1}^n f(x_k) \Delta x}{b-a}$

* Example 2: Use a finite sum to estimate the average value of f on the given interval into 4 subintervals of equal length and evaluating f at the subinterval midpoints.

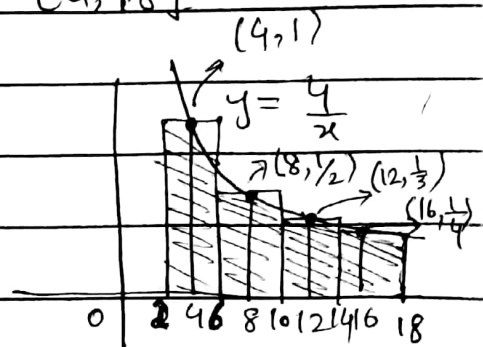
$$f(x) = \frac{4}{x} \text{ on } [2, 18]$$

Sol:

To find average value of $f(x)$ over $[2, 18]$, we have to find width Δx

$$\Delta x = \frac{18-2}{4} = 4$$

$$4 \quad 4$$



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$$\text{Midpoint} = \frac{a+b}{2} = \frac{2+6}{2} = \frac{8}{2} = 4.$$

$$4 \quad \frac{6+10}{2} = \frac{16}{2} = 8$$

$$4 \quad \frac{10+14}{2} = \frac{24}{2} = 12$$

$$4 \quad \frac{14+18}{2} = \frac{32}{2} = 16$$

$$\text{Area} = \sum_{k=1}^4 f(x_k) \Delta x$$

$$= f(x_1) \Delta x + f(x_2) \Delta x + f(x_3) \Delta x + f(x_4) \Delta x$$

$$= f(4) \Delta x + f(8) \Delta x + f(12) \Delta x + f(16) \Delta x$$

$$= 1 \cdot 4 + \frac{1}{2} \cdot 4 + \frac{1}{3} \cdot 4 + \frac{1}{4} \cdot 4$$

$$\text{Area} = \boxed{\frac{22}{3}}$$

$$\text{Average Value of } f(x) = \frac{\text{Area}}{b-a}$$

$$= \frac{22/3}{18-2}$$

$$= \frac{22/3}{16}$$

$$= \frac{22/3}{16}$$

$$\frac{16}{3} = 5.33$$

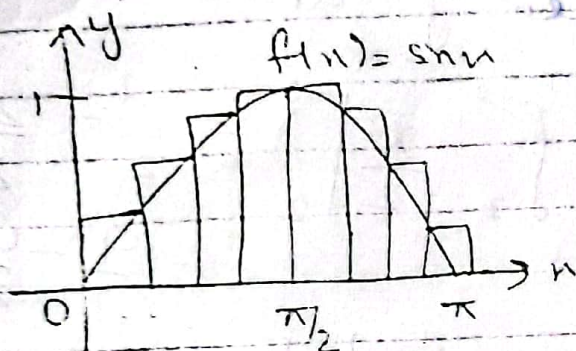
$$= \boxed{\frac{22}{48}}$$



→ Example #3

Estimate the average value of the function $f(x) = \sin x$ on the interval $[0, \pi]$.

Solution



$$\Delta x = \frac{b-a}{n} = \frac{\pi-0}{8} = \frac{\pi}{8}$$

→ To find the average, find area under graph and divide by $\pi-0 = \pi$.

→ Upper sum approximation → Add area of 8 rectangles of equal width $\pi/8$.

$$A \approx \left(\sin \frac{\pi}{8} + \sin \frac{\pi}{4} + \sin \frac{3\pi}{8} + \sin \frac{\pi}{2} + \sin \frac{\pi}{2} + \sin \frac{5\pi}{8} + \sin \frac{3\pi}{4} + \sin \frac{7\pi}{8} \right) \cdot \frac{\pi}{8} \approx 2.364$$

$$\text{Average value of } f(x) \approx \frac{2.364}{\pi} \approx 0.753 \text{ (Average)}$$